



Institute of Technology University of Moratuwa

Community-Based Ocean & Coastal Cleanup Management System
Project I Final Report

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Declaration

We declare that this thesis is our own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education. Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

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Dedication

This project is dedicated to our parents, lecturers, friends, and all individuals who continuously supported and encouraged us throughout the completion of this project. We also dedicate this work to environmental conservation efforts and communities working to protect marine ecosystems and coastal environments.

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Abstract

Ocean and coastal pollution has become a serious environmental problem that affects marine ecosystems, tourism, fisheries, and public health. Existing cleanup activities are often managed manually through social media and informal communication methods, creating difficulties in pollution reporting, verification, volunteer coordination, and project monitoring. To address these issues, this project proposes a Community-Based Ocean & Coastal Cleanup Management System as a Progressive Web Application (PWA) for managing pollution reporting and cleanup activities efficiently. The PWA supports installation on desktop and mobile devices, offline access to selected features, push notifications, and responsive user experiences without requiring installation from an app store.

The system allows registered users to submit pollution reports by providing descriptions, images, videos, and geographic location details. Community users can verify submitted reports through voting and commenting mechanisms to improve report reliability and reduce false reports. Administrators review suspicious reports and coordinate with government authorities for approval and official guidance. After approval the system creates cleanup projects. Then project owners can manage volunteers and divers, and monitor project progress. The administrator can generate analytics reports. The system also provides location-based alerts to notify nearby volunteers and divers about cleanup opportunities.

The proposed solution is developed using React.js for the frontend, Spring Boot for the backend, and MySQL for database management. The system follows a multi-tier architecture to ensure scalability, security, maintainability, and efficient communication between system components. The project aims to improve environmental awareness, increase community participation, and support organized ocean and coastal cleanup management activities effectively.

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List Of Abbreviations

Abbreviation Meaning

API	Application Programming Interface
UI	User Interface
UX	User Experience
DBMS	Database Management System
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
REST	Representational State Transfer
JSON	JavaScript Object Notation
NGO	Non-Governmental Organization
MEPA	Marine Environment Protection Authority

1. Introduction

1.1. Introduction

Marine pollution and coastal waste have become major environmental issues in Sri Lanka and many other countries. Ocean pollution affects marine ecosystems, tourism, fishing industries, and public health [1]. Although many volunteers and organizations participate in cleanup activities, there is no proper centralized system to report polluted locations, coordinate volunteers, verify reports, and monitor cleanup progress efficiently.

The proposed project, “Community-Based Ocean & Coastal Cleanup Management System,” is developed as a PWA to support pollution reporting, community verification, cleanup coordination, volunteer management, and project monitoring. This chapter introduces the background, motivation, objectives, and overall proposed solution of the system.

1.2. Background and Motivation

1.2.1. Background

Environmental pollution in coastal and marine areas is increasing rapidly due to plastic waste, industrial waste, tourism activities, and poor waste management practices [1]. Most cleanup activities are managed manually through social media posts, phone calls, or informal communication methods. These methods are inefficient, unorganized, and difficult to monitor.

Currently, there is no centralized digital platform that allows citizens, volunteers, organizations, and government authorities to work together effectively in managing coastal cleanup activities. Therefore, a proper management system is required to improve communication, reporting accuracy, volunteer coordination, and project monitoring.

1.2.2. Motivation

The following factors motivated the development of this project:

- Increasing ocean and coastal pollution problems.

- Lack of proper communication between volunteers and authorities.
- Difficulty in verifying pollution reports.
- Absence of centralized volunteer coordination.
- Need for real-time cleanup project monitoring.
- Requirement for better environmental awareness and community participation.
- Need for reliable analytics and reporting for decision-making.

1.3. Aim and Objectives

1.3.1. Aim

The main aim of this project is to develop a PWA, Community-Based Ocean & Coastal Cleanup Management System that supports pollution reporting, verification, volunteer coordination, cleanup project management, and environmental monitoring.

1.3.2. Objectives

The objectives of the project are:

- To allow users to report polluted coastal and ocean locations.
- To enable community verification for pollution reports.
- To support government authority approval processes.
- To coordinate volunteers and divers for cleanup projects.
- To provide location-based notifications and alerts.
- To monitor cleanup project progress using images and updates.
- To generate analytics and reports related to pollution incidents and cleanup activities.

- To provide a secure and user-friendly platform for all users.

1.4. Proposed Solution

The proposed solution is a PWA developed using React.js, Spring Boot, and MySQL. The system follows a multi-tier architecture consisting of presentation, application, and data layers.

The system allows users to submit pollution reports with images, videos, descriptions, and map locations. Community users can verify reports through voting and comments. Administrators review suspicious reports and coordinate with government authorities for approvals. Organizations can create cleanup projects and notify volunteers and divers nearby. The system also supports analytics, progress monitoring, and diver opportunity management.

1.4.1. Users

The Community-Based Ocean & Coastal Cleanup Management System is designed to support multiple categories of users, each with specific responsibilities and access privileges within the platform.

• Volunteer Non-Divers

Volunteer non-divers are registered community members who can report marine and coastal pollution incidents by submitting descriptions, images, videos, and location details. They can also participate in the community verification process by voting on and commenting on reported incidents.

• Volunteer Divers

Volunteer divers are trained individuals who participate in underwater cleanup activities.

In addition to reporting and verifying incidents, they can join cleanup projects that

require diving expertise, maintain diver profiles, and receive notifications about nearby underwater cleanup opportunities.

• Government Authority Officers

Government authority officers, such as representatives from the Marine Environment Protection Authority (MEPA), are responsible for reviewing cleanup requests, evaluating environmental concerns, providing official recommendations, and approving or rejecting cleanup activities when necessary.

- **Organizations**

Environmental organizations, NGOs, and community groups can create and manage cleanup projects through the system. They are responsible for coordinating volunteers, assigning tasks, monitoring project progress, and posting opportunities for divers and other participants.

- **System Administrators**

System administrators manage the overall operation of the platform. Their responsibilities include reviewing submitted reports, moderating user activities, managing accounts, handling suspicious or false reports, maintaining system integrity, and ensuring smooth platform operation.

- **Non-Registered Users**

Non-registered users can access publicly available information such as environmental awareness content, approved cleanup projects, project updates, and general statistics. However, they cannot submit reports or participate in system activities until they register for an account.

1.4.2. Input

The proposed system receives various types of data from users and external services to support pollution reporting, verification, project management, and monitoring activities.

The primary inputs to the system are as follows:

- **User Registration Details**

Information provided during account creation, including name, email address, contact information, user role, and password, which are used for user identification and authentication.

- **Login Credentials**

Username, email addresses, and passwords entered by registered users to securely access the system according to their assigned roles and permissions.

- **Pollution Report Information**

Details related to pollution incidents, including descriptions of the affected area, pollution type, severity level, and other relevant observations submitted by users.

- **Images and Videos**

Photographic and video evidence uploaded by users to support pollution reports and provide visual confirmation of environmental conditions.

- **Geographic Location Data**

Location information obtained through map selection or GPS coordinates to accurately identify polluted coastal or marine areas and facilitate cleanup planning.

- **Community Verification Data**

Votes, comments, feedback, and verification responses submitted by community members to assess the authenticity and reliability of reported incidents.

- **Cleanup Project Progress Information**

Project updates submitted by organizations or administrators, including completion percentages, progress notes, before-and-after photographs, and status reports.

- **Diver Profile Information**

Professional and personal details of volunteer divers, including certifications, skills, experience, availability, and preferred cleanup locations.

- **Government Approval and Review Data**

Approval decisions, comments, recommendations, and official instructions provided by government authorities during the cleanup approval process.

- **Analytics and Search Parameters**

Filtering criteria, search queries, date ranges, and report generation options selected by users to view statistics, trends, and analytical information within the system.

1.4.3. Output

The Community-Based Ocean & Coastal Cleanup Management System generates various outputs that support pollution management, volunteer coordination, project monitoring, and decision-making processes. The main outputs of the system are as follows:

- **Verified Pollution Reports**

Reports that have been validated through community verification and administrative review, providing reliable information about marine and coastal pollution incidents.

- **Cleanup Project Information**

Details of approved cleanup projects, including project objectives, locations, schedules, participating organizations, assigned volunteers, and project status.

- **Notifications and Alerts**

Location-based notifications and system alerts sent to volunteers, divers, organizations, and administrators regarding new pollution incidents, cleanup opportunities, project updates, and important announcements.

- **Project Progress Updates**

Progress reports containing status updates, completion percentages, photographs, and activity summaries that help stakeholders monitor cleanup operations.

- **Government Approval Status**

Approval decisions, recommendations, and official comments provided by government authorities regarding proposed cleanup activities and environmental concerns.

- **Analytics Dashboards and Reports**

Visual dashboards, charts, and reports that present pollution statistics, cleanup performance indicators, volunteer engagement levels, and environmental trends.

- **Volunteer and Diver Participation Statistics**

Information related to volunteer registrations, participation rates, completed cleanup activities, and diver involvement in environmental conservation projects.

- **Regional Pollution Trends**

Analytical reports that identify pollution hotspots, frequently affected regions, and longterm environmental patterns to support planning and decision-making.

- **Environmental Awareness Information**

Publicly accessible information, project outcomes, success stories, and awareness content that encourage community participation in environmental conservation efforts.

1.4.4. Process

The process of the Community-Based Ocean & Coastal Cleanup Management System begins when users register and create accounts according to their respective roles, such as volunteers, divers, organizations, government officers, or administrators. Once registered, community users can report polluted coastal or marine locations by providing descriptions, images, videos, and geographic location details.

After a report is submitted, it becomes available for community verification. Registered users can review the report, vote on its authenticity, and provide comments. Based on these responses, the system calculates a trust percentage to determine whether the report can be considered community verified. Reports that fail to achieve the required verification threshold are forwarded to the administrator for further review.

The administrator evaluates submitted reports, reviews evidence and community feedback, and may approve, reject, or request additional clarification from the report submitter. Verified and approved reports are then forwarded to the relevant government authority for official review and approval. Government officers can provide approval decisions, recommendations, or instructions related to the cleanup activity.

Once the required approvals are obtained, the system then creates a cleanup project based on the approved reports. Then the system identifies nearby volunteers and divers and sends location-based notifications inviting them to participate. Users may accept or decline participation requests, and the system records volunteer involvement.

During project execution, authorized users can upload progress updates, photographs, and completion information to monitor project activities. The system maintains a progress timeline and records cleanup outcomes for transparency and accountability.

In addition, organizations can publish diver-related opportunities, allowing qualified divers to showcase their skills and participate in environmental conservation activities. Finally, the system collects and analyses operational data to generate reports, participation statistics, pollution trends, and analytical dashboards that support environmental monitoring and decision-making.

1.5. Summary

This chapter introduced the Community-Based Ocean & Coastal Cleanup Management System by discussing the environmental challenges of marine pollution, the motivation for developing the system, and its aims and objectives. It also provided an overview of the proposed solution, including its users, inputs, outputs, and overall operational process. These aspects establish the foundation of the project and demonstrate how the system supports effective pollution reporting, cleanup coordination, and environmental conservation. The next chapter reviews existing approaches and related systems relevant to the proposed solution.

2. Literature Review

2.1. Introduction

This chapter reviews existing approaches and technologies related to environmental pollution reporting, volunteer coordination, and cleanup management systems. The purpose of this review is to identify the strengths and limitations of existing solutions and compare them with the proposed system.

2.2. Available Approaches

1. Marine Debris Tracker

Marine Debris Tracker is a mobile application used to record marine debris locations. Users can submit pollution information with geographic coordinates. The system mainly focuses on data collection and environmental research [3].

Features:

- Pollution reporting
- GPS location tracking
- Data collection and visualization

Limitations:

- No volunteer coordination
- No government approval workflow
- No cleanup project monitoring



Figure 1: Marine Debris Tracker Interface [3]

<https://debristracker.org/>

2. Clean Swell

Developed by Ocean Conservancy, Clean Swell allows volunteers to record trash collected during beach and environmental cleanup activities. The application tracks cleanup locations using GPS, records collected waste data and contributes information to a global marine debris database. However, it does not provide community verification, government approval workflows, or project management capabilities [4].

Features

- Cleanup activity recording
- GPS tracking
- Waste data collection
- Global environmental database

Limitations

- No community-based report verification
- No government authority integration
- No volunteer project coordination
- No cleanup approval workflow



Figure 2: Clean Swell System [4]

<https://oceanconservancy.org/work/plastics/cleanups-icc/clean-swell-app/>

3. MEPA Beach Caretaker Program

MEPA operates a Beach Caretaker program where designated personnel identify polluted beaches and carry out cleaning activities. While effective for beach maintenance, the program

does not provide public reporting or volunteer coordination through a digital system [2].

Features

- Beach monitoring
- Organized cleanup activities

Limitations

- No citizen reporting
- No volunteer management
- No digital workflow

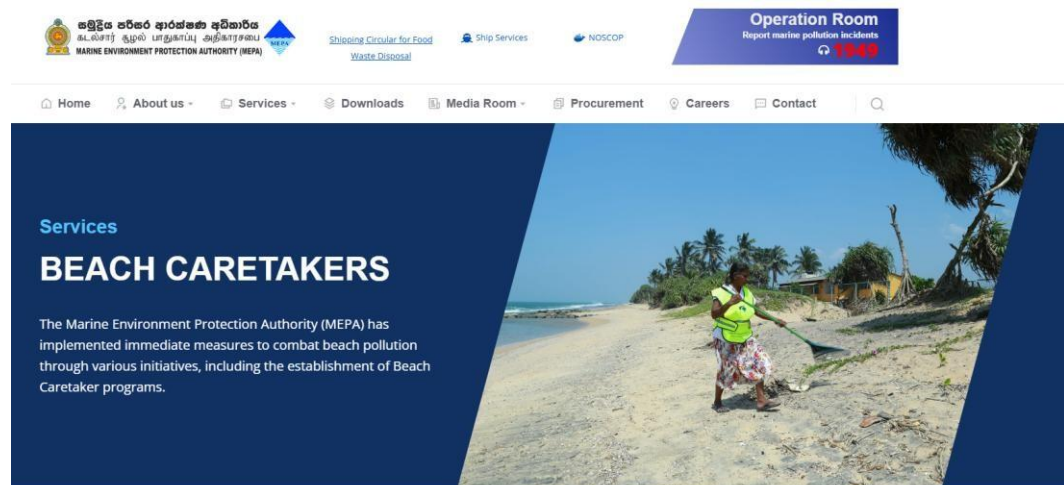


Figure 3: MEPA Beach caretaker Website [2]

<https://mepa.gov.lk/services/beach-caretakers/>

4. Parley Sri Lanka

Parley Sri Lanka conducts marine conservation and cleanup activities with volunteers and partner organizations. Their focus is on environmental action and awareness rather than providing a complete cleanup management platform [5].

Features:

- Volunteer mobilization
- Beach cleanups
- Environmental awareness

Limitations:

- No pollution reporting system
- No government approval workflow
- No project monitoring dashboard



**PLASTIC IS A DESIGN FAILURE. FIXING IT IS
OUR DESIGN CHALLENGE**

Figure 4: Parley Sri Lanka Website [5]

<https://parley.tv/sri-lanka>

5. Social Media-Based Reporting

Many communities use Facebook groups and WhatsApp groups to report pollution incidents and organize cleanup activities.

Features:

- Fast communication
- Public awareness
- Easy information sharing

Limitations:

- No structured data management
- Difficult to verify reports
- No analytics or tracking
- Poor coordination

2.3. Summary

The literature review shows that existing systems provide only partial solutions for marine pollution management. Most available systems focus either on pollution reporting or volunteer management separately. However, they lack complete integration of reporting, verification, authority coordination, project monitoring, and analytics.

The proposed Community-Based Ocean & Coastal Cleanup Management System combines all major features into a single platform. It supports pollution reporting, community verification, government approval, volunteer coordination, diver opportunities, location-based alerts, project monitoring, and analytics within one centralized system.

Features	Marine Debris Tracker	Clean Swell	MEPA Beach Caretaker	Parley Sri Lanka	Social Media Reporting	Proposed System
Pollution Reporting	√	X	X	X	√	√
Community Verification	X	X	X	X	X	√
Government Coordination	X	X	X	X	X	√

Volunteer Coordination	X	X	Limited	√	Limited	√
Diver Management	X	X	X	X	X	√
Project Monitoring	X	X	Limited	X	X	√
Location- Based Alerts	X	√	X	X	X	√
Analytics Dashboard	√	Limited	X	X	X	√

Table 1: Feature Comparison Between Proposed Solution and Existing Systems

3. Approach

3.1. Introduction

This chapter explains the overall approach and architecture used in the proposed Community-Based Ocean & Coastal Cleanup Management System. It also describes the main modules and functionalities of the system.

3.2. Approach

The proposed system follows a multi-tier architecture consisting of the following layers:

- Presentation Layer
- Application Layer
- Data Layer
- External Service Layer

The frontend is implemented as a PWA using React.js. The backend is developed using Spring Boot to handle business logic and REST API services.

MySQL is used for permanent data storage.

3.2.1. User Registration and Profile Management Module

This module manages user registration, authentication, and profile maintenance within the system. It allows different categories of users, including volunteers, divers, organizations, government officers, and administrators, to create accounts and access system functionalities according to their assigned roles and permissions. The module ensures secure user access and supports the management of personal information and user-specific activities.

Functions:

- User registration
- Login authentication
- Role-based access control
- Profile management

3.2.2. Pollution Reporting Module

The Pollution Reporting Module serves as the primary entry point for reporting marine and coastal pollution incidents. Registered users can submit detailed reports by providing descriptions, images, videos, and geographic location information. The collected information is stored in the system and forwarded for community verification and further review, helping create a centralized repository of environmental incidents.

Functions:

- Submit pollution reports
- Upload evidence
- Capture location details
- Store report information

3.2.3. Community Verification Module

This module enables community participation in verifying reported pollution incidents. Users can review submitted reports, provide comments, and vote on their authenticity. Based on the feedback received, the system calculates a trust percentage that helps determine whether a report should proceed to the next stage of review. This process improves report reliability and reduces false or misleading submissions.

Functions:

- Vote true or false

- Add comments
- Calculate trust percentage
- Mark verified reports

3.2.4. Admin Review Module

The Admin Review Module provides administrators with the tools required to manage reports and maintain the integrity of the platform. Administrators can review suspicious or unverified reports, evaluate supporting evidence, approve or reject submissions, and request additional information when necessary. The module also supports user management and content moderation activities.

Functions:

- Review reports
- Approve or reject requests
- Manage users
- Submit Clarification Request

3.2.5. Government Coordination Module

This module facilitates communication between the system and relevant government authorities responsible for environmental protection and cleanup approvals. Approved reports can be forwarded for official review, allowing authorities to provide decisions, recommendations, and guidelines. The module ensures that cleanup activities comply with environmental regulations and approval requirements.

Functions:

- Forward reports
- Review approvals

- Provide authority comments
- Maintain approval status

3.2.6. Alert and Notification Module

The Alert and Notification Module is responsible for informing users about relevant events and activities within the system. Using location-based services, the module identifies nearby volunteers and divers and sends notifications regarding pollution incidents, cleanup opportunities, project updates, and important announcements. This helps improve participation and response rates for cleanup activities.

Functions:

- Location-based alerts
- Project notifications
- User responses
- Alert expansion

3.2.7. Project Monitoring Module

This module supports the monitoring and management of cleanup projects from initiation to completion. Organizations and administrators can record project progress through status updates, photographs, completion percentages, and activity reports. The module provides transparency and enables stakeholders to track the effectiveness and outcomes of cleanup operations.

Functions:

- Upload progress updates
- Monitor project status
- Display timelines
- Store cleanup results

3.2.8. Analytics Module

The Analytics Module provides reporting and visualization capabilities for monitoring pollution incidents, cleanup activities, volunteer participation, and environmental trends. It assists administrators and organizations in making data-driven decisions through dashboards and statistical reports. Functions:

- Generate reports
- Display dashboards
- Show regional trends
- Analyse participation data

3.2.9. Diver Opportunity Management Module

This module connects qualified divers with organizations conducting underwater cleanup activities. Organizations can publish diver opportunities and search for suitable participants based on their skills and experience. Divers can maintain professional profiles, showcase certifications, and apply for relevant cleanup projects. The module helps improve the utilization of specialized diving expertise in marine conservation efforts.

Functions:

- Create diver opportunities
- Manage diver profiles
- Search qualified divers
- Apply for cleanup opportunities

3.3. Summary

This chapter discussed the overall approach and architecture of the proposed CommunityBased Ocean & Coastal Cleanup Management System. It explained the multi-tier architecture adopted for the system and described the major functional modules responsible for user management, pollution reporting, verification, approval workflows, volunteer coordination, project monitoring, diver opportunity management, and analytics. Together, these modules provide an integrated platform for managing ocean and coastal cleanup activities effectively. The next chapter discusses the technologies and tools selected for the implementation of the proposed solution.

4. Technology

4.1. Introduction

This chapter discusses the technologies, frameworks, development tools, and software platforms selected for implementing the Community-Based Ocean & Coastal Cleanup Management System. The chosen technologies support the development of a secure, scalable, and maintainable Progressive Web Application while providing efficient data management, responsive user interfaces, and reliable system performance.

4.2. Technologies Adapted

4.2.1. Programming Languages

The system uses multiple programming languages for frontend and backend development.

- **JavaScript**

JavaScript is used as the primary programming language for frontend development using React.js. It enables the creation of dynamic and interactive user interfaces, supports client-side validation, handles user interactions, and facilitates communication with backend services through API requests. JavaScript contributes significantly to improving the responsiveness and usability of the web application [6].

- **Java**

Java is used for backend development through the Spring Boot framework. It is responsible for implementing business logic, processing user requests, managing authentication and authorization, handling data transactions, and providing RESTful API services. Java was selected due to its reliability, scalability, and extensive support for enterprise-level application development [8].

- **SQL**

SQL was selected as the standard database query language because it provides a reliable and efficient way to create, retrieve, update, and manage structured data within relational databases. SQL was chosen due to its widespread adoption, compatibility with MySQL, support for complex queries, data integrity, and efficient transaction management. It enables the proposed system to manage user information, pollution reports, project data, notifications, and analytics efficiently while maintaining consistency and accuracy of stored data.

4.2.2. Libraries and Frameworks

- **React.js**

React.js is a modern JavaScript library used for developing the frontend of the system. It follows a component-based architecture that allows developers to build reusable and maintainable user interface components. React improves application performance through efficient rendering mechanisms and supports the development of responsive web interfaces [6].

Advantages:

- Fast rendering and improved performance
- Reusable UI components
- Enhanced user experience
- Responsive interface design
- Simplified application maintenance
- Supports Progressive Web Application development
- Easy integration with Service Workers
- Supports installable web applications

- **Spring Boot**

Spring Boot is used as the primary backend framework for the proposed system. It simplifies application development by providing built-in configurations, dependency management, and integration support. Spring Boot enables the creation of secure and scalable RESTful web services that support the core functionalities of the system [7].

Advantages:

- Simplified application development
- RESTful API support
- Built-in dependency management
- Security integration
- Easy database connectivity
- Scalable architecture

4.2.3. Progressive Web Application (PWA)

Progressive Web Application (PWA) is a modern web application approach that combines the accessibility of traditional websites with the functionality and user experience of native mobile applications. The proposed Community-Based Ocean & Coastal Cleanup Management System adopts the PWA approach to provide users with a fast, reliable, and accessible platform across both desktop and mobile devices. Since volunteers, community members, and government authorities may need to access the system in various locations where internet connectivity can be limited or unstable, a PWA enables essential application resources to remain available through offline caching while improving overall performance. Additionally, users can install the application directly from their web browser without requiring an app store, making the system easier to access and encouraging greater community participation.

Advantages:

- Installable on Android/Desktop

- Offline functionality
- Push notifications
- Faster loading
- Cross-platform
- No app store required

- **Service Worker**

A Service Worker is a background script that runs independently of the web application and enables key Progressive Web Application features such as resource caching and offline access. It was used in the proposed system to improve application performance by storing frequently accessed resources locally, allowing pages to load faster and remain accessible even when network connectivity is limited or temporarily unavailable. This is particularly beneficial for volunteers and community members who may access the system in remote coastal areas with unstable internet connections.

- **Web App Manifest**

The Web App Manifest is a configuration file that defines how the Progressive Web Application appears and behaves when installed on a user's device. It specifies information such as the application name, icons, theme color, display mode, and start URL. In the proposed system, the Web App Manifest enables users to install the application directly from their web browser, providing a native app-like experience on both desktop and mobile devices without requiring installation through an application store.

4.2.4. RESTful APIs

RESTful APIs are used to facilitate communication between the React.js frontend and the Spring Boot backend. The APIs enable secure data exchange for user authentication, pollution reporting, project management, verification processes, and analytics generation.

Advantages:

- Standardized communication mechanism
- Platform independence
- Scalability and flexibility
- Easy integration with external services

4.2.5. Integrated Development Environments (IDEs)

Several development tools were used throughout the project implementation process to improve productivity and code management.

- Visual Studio Code – Used for frontend development, UI implementation, and JavaScript programming.
- IntelliJ IDEA / Spring Tool Suite (STS) – Used for backend development, API implementation, and Spring Boot project management.
- MySQL Workbench – Used for database design, query execution, and database administration tasks.

4.2.6. Database / Data Storage

○ MySQL

MySQL was selected as the primary relational database management system because it offers high performance, reliability, scalability, and seamless integration with the Spring Boot framework. It

provides efficient storage and retrieval of structured data while ensuring data consistency and security. MySQL is also open source, widely used in enterprise and web applications, and supported by a large developer community, making it a suitable database solution for the proposed Progressive Web Application. [9].

The database stores:

- User information
- Pollution reports
- Images and videos
- Verification records
- Project details
- Volunteer information
- Notifications
- Analytics data

Advantages:

- Reliable performance
- Structured data management
- Data integrity and security
- Efficient query processing
- Easy integration with Spring Boot

4.2.7. Google Maps API

Google Maps API is used to support location-based functionalities within the system. It enables users to identify polluted locations, view cleanup project locations, and provide accurate geographic coordinates when submitting reports [10].

Advantages:

- Accurate location services
- Interactive maps
- GPS coordinate integration
- Improved user experience

4.2.8. Docker

Docker is used to containerize application components and provide a consistent deployment environment. It simplifies application setup, improves portability, and ensures that the system behaves consistently across development and production environments [11].

Advantages:

- Simplified deployment
- Environment consistency
- Scalability
- Easier maintenance

4.2.9. Version Control and Build Tools

Git is used as the version control system for tracking source code changes throughout the development lifecycle. GitHub serves as the remote repository platform that enables collaboration among team members, code sharing, version management, and project documentation.

Advantages:

- Source code backup and recovery
- Team collaboration
- Version tracking and history management
- Branch management
- Efficient project coordination

○ Maven

Maven is used as the build automation and dependency management tool for the Spring Boot backend application. It simplifies project configuration, dependency handling, compilation, testing, and packaging processes, allowing developers to manage the application efficiently [12].

4.2.10. Application and Web Servers

○ Apache Tomcat

Apache Tomcat is used as the embedded web server provided by Spring Boot. It hosts backend services and REST APIs while handling incoming HTTP requests and responses. Tomcat enables efficient communication between the frontend application and backend services and supports reliable application deployment [13].

4.2.11. Cloud Computing

The proposed system is designed to support deployment on cloud computing platforms such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure. Cloud-based deployment provides flexibility, scalability, and improved accessibility while supporting future system expansion and maintenance.

Benefits:

- Improved scalability
- High availability
- Remote accessibility
- Data backup and recovery support
- Enhanced deployment flexibility

4.2.12. UI, UX and Graphic Tools

Several design and visualization tools were used during the planning and development stages of the project.

- Figma – Used for designing user interfaces, wireframes, and user experience prototypes.
- Canva – Used for creating presentations, graphical content, and project-related visual materials.
- Draw.io – Used for developing system diagrams, architecture diagrams, flowcharts, and other technical documentation.

4.3. Summary

This chapter presented the technologies, frameworks, development tools, and supporting services used in the implementation of the Community-Based Ocean & Coastal Cleanup Management System. Technologies such as React.js, Spring Boot, MySQL, Git, Docker, and Google Maps API were selected to ensure scalability, maintainability, security, and efficient system performance. Together, these technologies provide the foundation required to support the functional and non-functional requirements of the proposed solution. The next chapter discusses the requirement gathering process and system analysis.

5. Analysis

5.1. Introduction

This chapter presents the requirement gathering and analysis activities carried out for the development of the Community-Based Ocean & Coastal Cleanup Management System. It describes the techniques used to collect information from stakeholders, existing systems, and relevant sources, as well as the process used to analyze and organize the collected data. The chapter also discusses the functional and non-functional requirements identified for the proposed solution, which form the foundation for the system design and implementation.

5.2. Requirement Gathering and Analysis

To ensure that the proposed system addresses real-world environmental and operational challenges, several requirement gathering techniques were used during the analysis phase. Information collected from stakeholders, existing systems, and relevant literature was analysed to identify the functional and non-functional requirements of the Community-Based Ocean & Coastal Cleanup Management System.

5.2.1. Stakeholder Interviews

Requirement gathering interviews were conducted to understand the current processes, challenges, and expectations related to marine pollution reporting and cleanup management activities. Particular attention was given to the operational procedures followed by environmental authorities and community organizations. The interviews helped identify requirements related to pollution reporting, report verification, cleanup approvals, volunteer coordination, safety considerations, and project monitoring.

5.2.2. Existing System Study

Several existing environmental reporting and cleanup management platforms were studied, including Marine Debris Tracker, Clean Swell, and other community-based reporting approaches. The study focused on identifying useful features, understanding system limitations, and determining opportunities for improvement. The findings helped define the unique features of the proposed solution.

5.2.3. Literature Review and Online Research

Research articles, environmental conservation resources, government publications, and marine pollution management references were reviewed to gain a deeper understanding of environmental challenges and best practices in cleanup management. This research provided valuable insights into pollution reporting mechanisms, volunteer engagement strategies, and environmental monitoring approaches.

5.2.4. Requirement Analysis

After collecting the required information, the data was analysed and organized into functional and non-functional requirements. The analysis focused on identifying stakeholder needs, system functionalities, security requirements, performance expectations, usability considerations, scalability requirements, and reliability factors. These requirements were subsequently used as the foundation for the design and development of the proposed system.

5.3. Requirements of the Suggested Solution

Based on the requirement gathering and analysis activities, a comprehensive set of functional and non-functional requirements was identified for the Community-Based Ocean & Coastal Cleanup Management System. These requirements define the services, features, quality attributes, and operational characteristics that the system must satisfy to achieve its objectives.

5.3.1. Functional Requirements

Functional requirements describe the specific functionalities and services that the system must provide to its users.

Requirement ID	Functional Requirement
FR-01	The system shall allow users to register and create accounts according to their respective roles.
FR-02	The system shall allow registered users to securely log in and access authorized functionalities.
FR-03	The system shall support role-based access control for volunteers, divers, organizations, government officers, and administrators.
FR-04	The system shall allow users to create and manage personal profiles.
FR-05	The system shall enable users to submit pollution reports with descriptions, images, videos, and location details.
FR-06	The system shall support map-based location selection and GPS coordinate capture.
FR-07	The system shall allow community members to verify reports through voting and commenting mechanisms.
FR-08	The system shall calculate trust percentages to determine report credibility.
FR-09	The system shall allow administrators to review, approve, reject, or request clarification for submitted reports.
FR-10	The system shall support government authority review and approval workflows.

FR-11	The system shall send location-based notifications and alerts to nearby volunteers and divers.
FR-12	The system shall allow organizations to create, manage, and monitor cleanup projects.
FR-13	The system shall support volunteer and diver participation in cleanup activities.
FR-14	The system shall maintain progress updates, timelines, and completion records for cleanup projects.
FR-15	The system shall allow organizations to create and manage diver opportunities.
FR-16	The system shall generate analytical reports, dashboards, and environmental statistics.
FR-17	The system shall provide users with access to environmental awareness information and project updates.

Table 2: Functional Requirements

5.3.2. Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational standards that the system must satisfy.

Performance Requirements:

Requirement ID	Non-Functional Requirement
NFR-01	The system should load web pages within 3 to 5 seconds under normal operating conditions.

NFR-02	The system should support concurrent access by multiple users without significant performance degradation.
NFR-03	Notifications, alerts, and system responses should be delivered efficiently and reliably.
NFR-04	The system shall cache frequently used resources to improve performance and limited offline access.

Table 3: Performance Requirements

Security Requirements:

Requirement ID	Non-Functional Requirement
NFR-05	User passwords shall be encrypted before storage.
NFR-06	HTTPS shall be used to secure communication between system components.
NFR-07	Role-based access control shall restrict access to authorized users only.
NFR-08	The system shall protect sensitive information from unauthorized access and modification.

Table 4: Security Requirements

Usability Requirements:

Requirement ID	Non-Functional Requirement
NFR-09	The user interface should be intuitive, user-friendly, and easy to navigate.
NFR-10	The system should support both desktop and mobile web browsers.

NFR-11	Users should receive clear feedback, notifications, and error messages during system interactions.
NFR-12	The system shall support installation as a Progressive Web Application on supported browsers.
NFR-13	The system shall support push notifications for project alerts and cleanup activities.
NFR-14	The application shall automatically synchronize offline data once an internet connection is restored.

Table 5: Usability Requirements

Reliability Requirements:

Requirement ID	Non-Functional Requirement
NFR-15	The system should maintain stable and consistent operation during normal usage.
NFR-16	Backup and recovery mechanisms shall be available to prevent data loss.
NFR-17	The system shall ensure data consistency and integrity throughout all transactions.

Table 6: Reliability Requirements

Scalability Requirements:

Requirement ID	Non-Functional Requirement
NFR-18	The system should support future feature enhancements and functional expansion.

NFR-19	The architecture should accommodate increasing numbers of users, reports, and cleanup projects without major redesign.
NFR-20	The system should support integration with additional environmental and government services in future versions.

Table 7: Scalability Requirements

5.4. Summary

This chapter presented the requirement gathering and analysis activities carried out for the proposed system. Various techniques, including stakeholder interviews, existing system studies, and literature research, were used to identify the needs and expectations of users and other stakeholders. Based on the collected information, a comprehensive set of functional and non-functional requirements was defined to guide the design and implementation of the Community-Based Ocean & Coastal Cleanup Management System. These requirements establish the foundation for developing a secure, reliable, scalable, and user-friendly solution. The next chapter discusses the design of the proposed system, including its architecture, database structure, and major design components.

6. Design

6.1. Introduction

This chapter presents the design of the Community-Based Ocean & Coastal Cleanup Management System. It describes the architectural design, functional design, database design, and user interface design of the proposed solution. The diagrams and models presented in this chapter illustrate how the different system components interact to support pollution reporting, verification, volunteer coordination, government approval, project monitoring, and analytics generation.

6.2. Design

6.2.1. Use Case Diagram Design

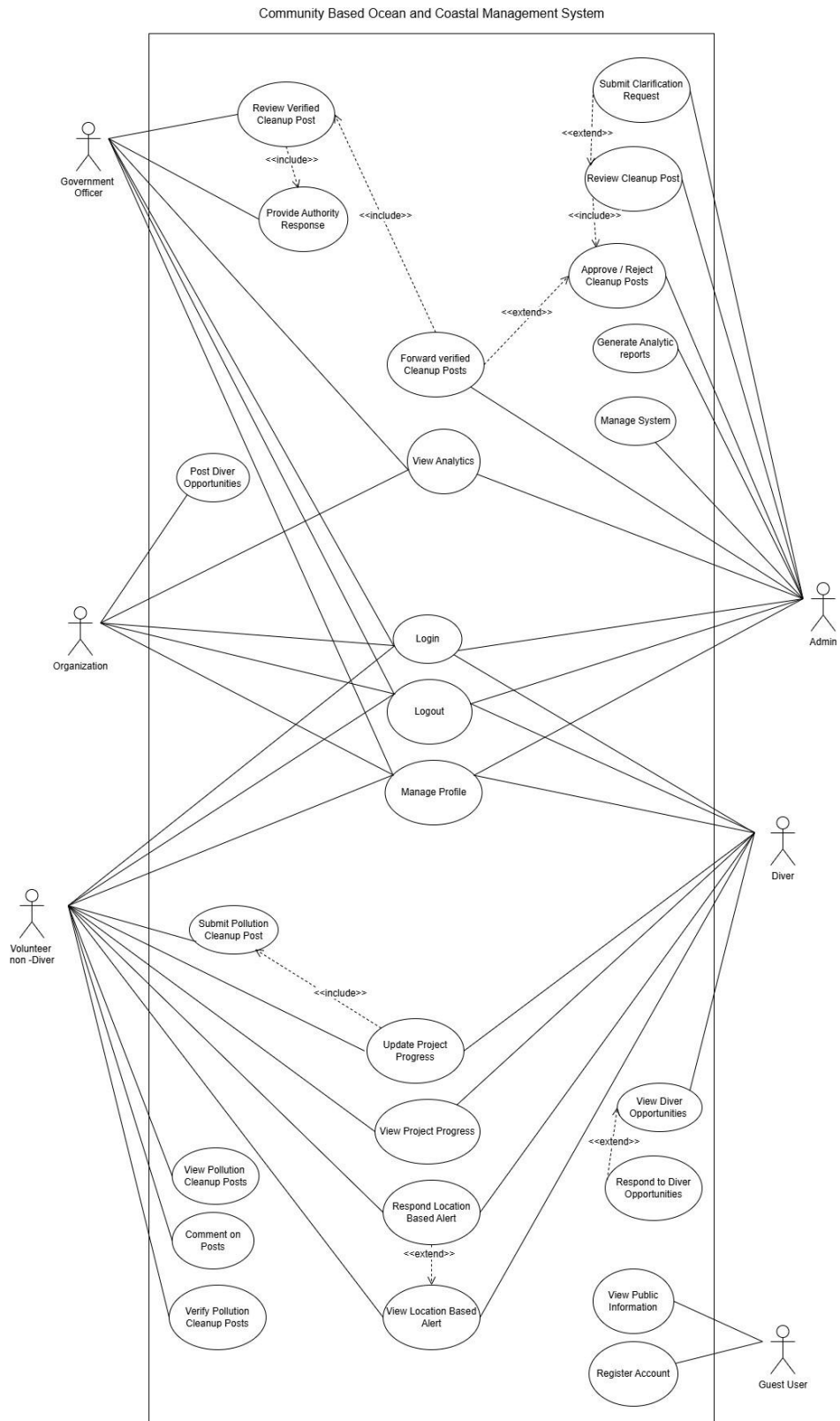


Figure 5: Use case Diagram

The Use Case Diagram illustrates the interactions between the different user roles and the Community-Based Ocean & Coastal Cleanup Management System. The system supports six primary actors: Volunteer Non-Divers, Divers, Organizations, Government Officers, Administrators, and Guest Users. Each actor is provided with specific functionalities based on their responsibilities and access privileges.

Volunteer Non-divers can register and log in to the system, manage their profiles, submit pollution cleanup requests, verify reported incidents, comment on reports, respond to location-based alerts, and monitor project progress. Divers have similar capabilities and can additionally view and respond to diver-specific opportunities published by organizations. Organizations can manage their profiles, view analytics, post diver opportunities, and participate in cleanup project coordination activities.

Government Officers are responsible for reviewing verified cleanup requests forwarded by administrators. They can evaluate request details, provide official responses, and approve or reject proposed cleanup activities. Administrators have the highest level of system access and are responsible for reviewing cleanup requests, approving or rejecting reports, requesting clarifications, managing users, maintaining system operations, and generating analytical reports. Guest Users can access publicly available information and register for an account to gain access to additional system functionalities.

The relationships shown in the diagram also illustrate the dependencies between use cases. For example, government officers review verified cleanup requests before providing official responses, while administrators review reports before making approval decisions. These interactions collectively support the complete workflow of pollution reporting, verification, approval, volunteer coordination, project monitoring, and environmental management within the proposed system.

6.2.2. Activity Diagrams Design

Pollution Cleanup Post Submission

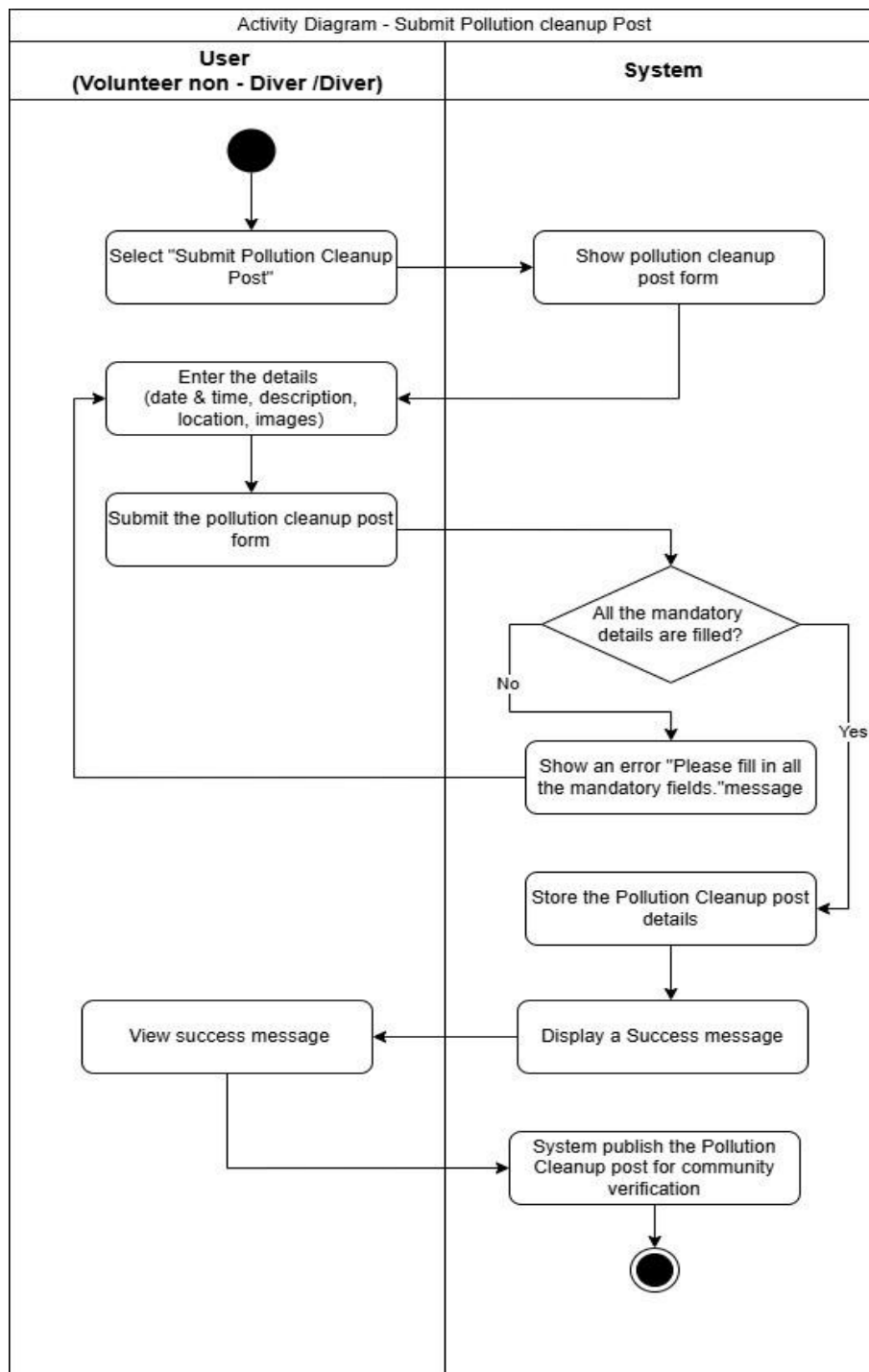


Figure 6: Activity Diagram - Pollution Cleanup Post Submission

This activity diagram illustrates the process of submitting a pollution cleanup post to the system. The process begins when a volunteer non-diver or diver selects the "Submit Pollution Cleanup Post" option. The system then displays a reporting form where the user enters the required information, including the date and time of the incident, a description of the pollution, location details, and supporting images.

Once the form is submitted, the system validates the entered information to ensure that all mandatory fields have been completed. If any required information is missing, an error message is displayed, and the user is prompted to provide the missing details. If the validation is successful, the system stores the pollution report, displays a confirmation message, and publishes the report for community verification. This process ensures that pollution incidents are properly documented and made available for further review and verification by the community.

Community Verification

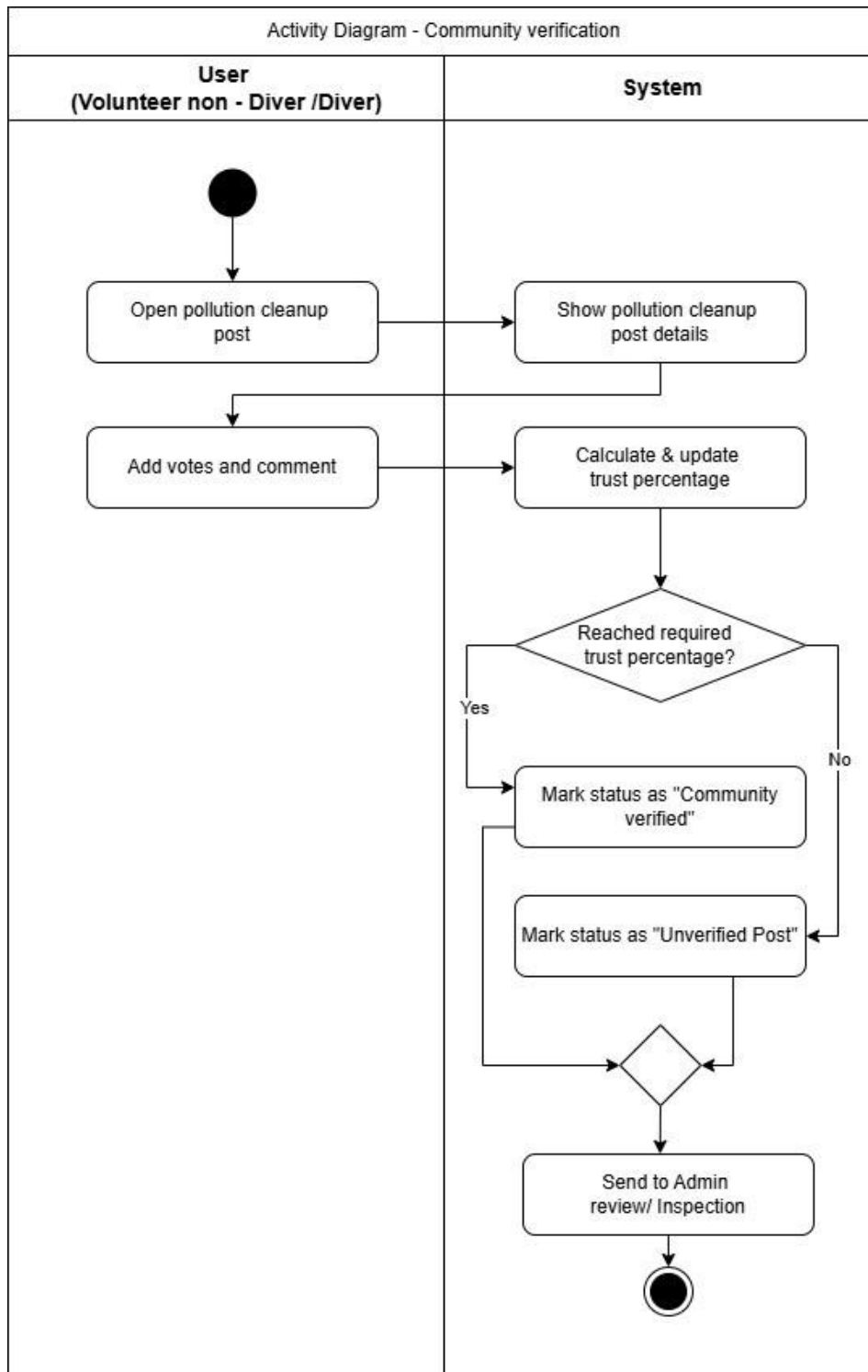


Figure 7: Activity Diagram - Community Verification

This activity diagram illustrates the community verification process used to assess the credibility of submitted pollution cleanup posts. The process begins when a volunteer non-diver or diver opens a pollution cleanup post. The system displays the details of the reported incident, allowing community members to review the information and supporting evidence before providing their feedback.

Users can then submit votes and comments regarding the authenticity of the report. Based on the feedback received, the system calculates and updates the trust percentage associated with the post. If the required trust percentage is achieved, the report is marked as "Community Verified." Otherwise, it is marked as an "Unverified Post." Regardless of the verification outcome, the report is forwarded to the administrator for further review and inspection. This process promotes community participation while improving the reliability and accuracy of pollution reports submitted to the system.

Admin Review Process

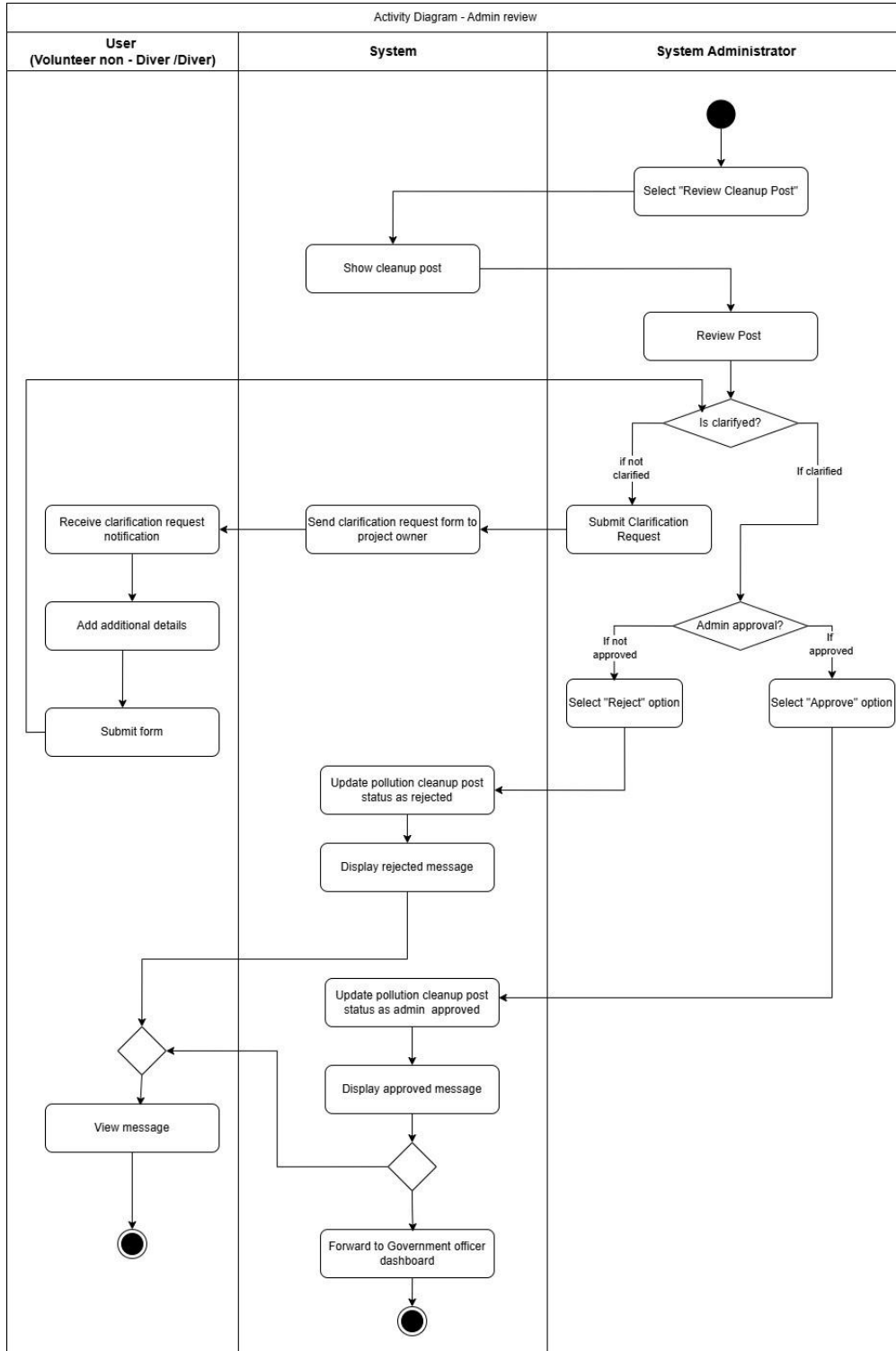


Figure 8: Activity Diagram - Admin Review Process

This activity diagram illustrates the administrative review process for pollution cleanup posts. The administrator reviews the submitted report and determines whether additional clarification is required. If further information is needed, a clarification request is sent to the report owner, who provides the requested details and resubmits the report. Once the report is sufficiently reviewed, the administrator either approves or rejects it. Rejected reports are updated accordingly and the user is notified, while approved reports are forwarded to the government officer dashboard for official review and approval.

Government Officer Review Process

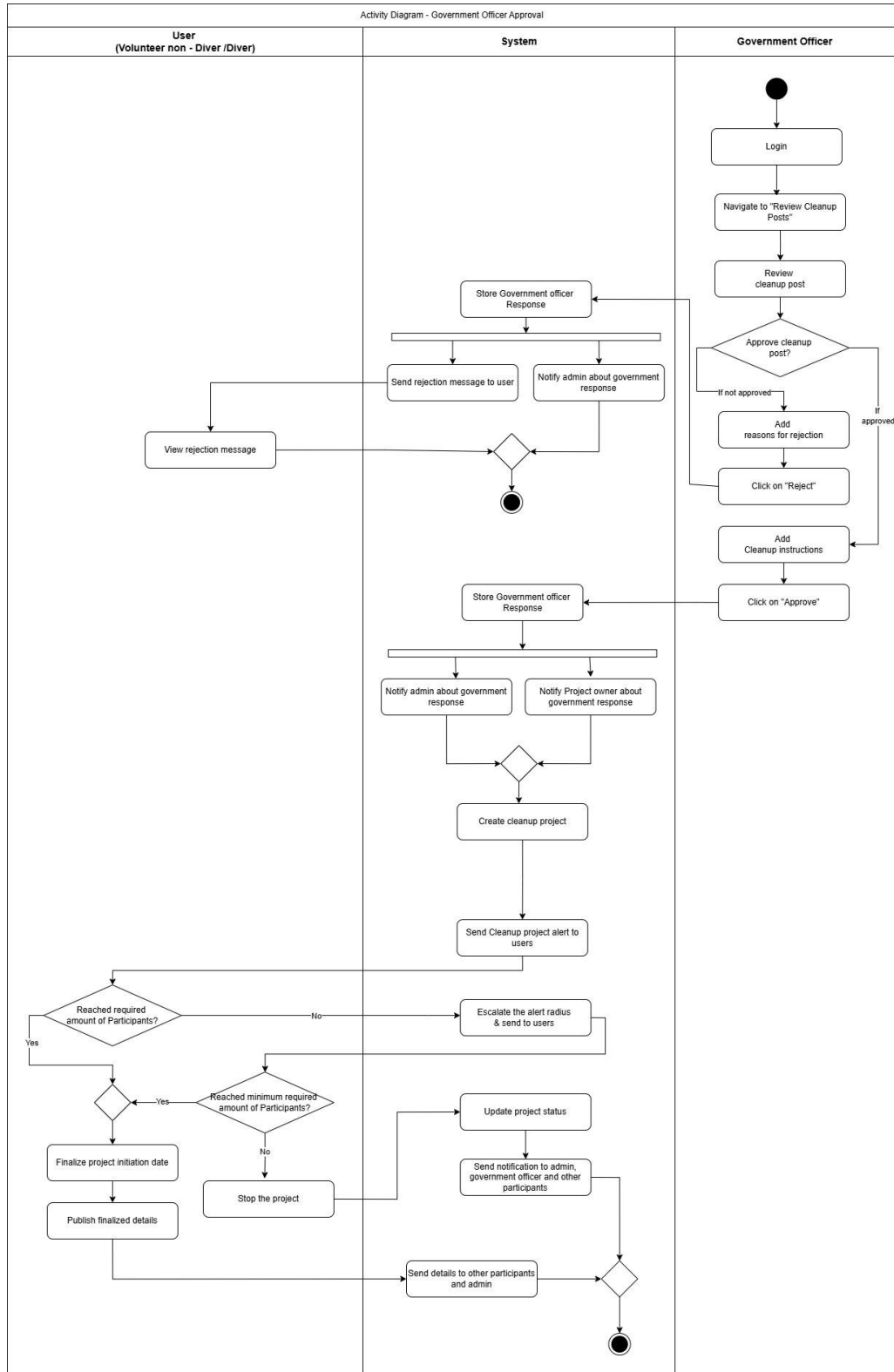


Figure 9: Activity Diagram - Government Officer Review Process

This activity diagram illustrates the government approval process for pollution cleanup posts that have been approved by the administrator. The government officer reviews the cleanup request and either approves or rejects it based on the information provided. If the request is rejected, the reasons for rejection are recorded and relevant parties are notified. If approved, cleanup instructions are provided and a cleanup project is created. The system then notifies potential participants and monitors registrations. If sufficient participants join, the project details are finalized and published; otherwise, the notification radius is expanded to attract additional participants, and the project may be cancelled if the minimum participation requirement is not met. Throughout the process, the system updates project status information and notifies administrators, government officers, and participants of important developments.

Alert and Notification Process

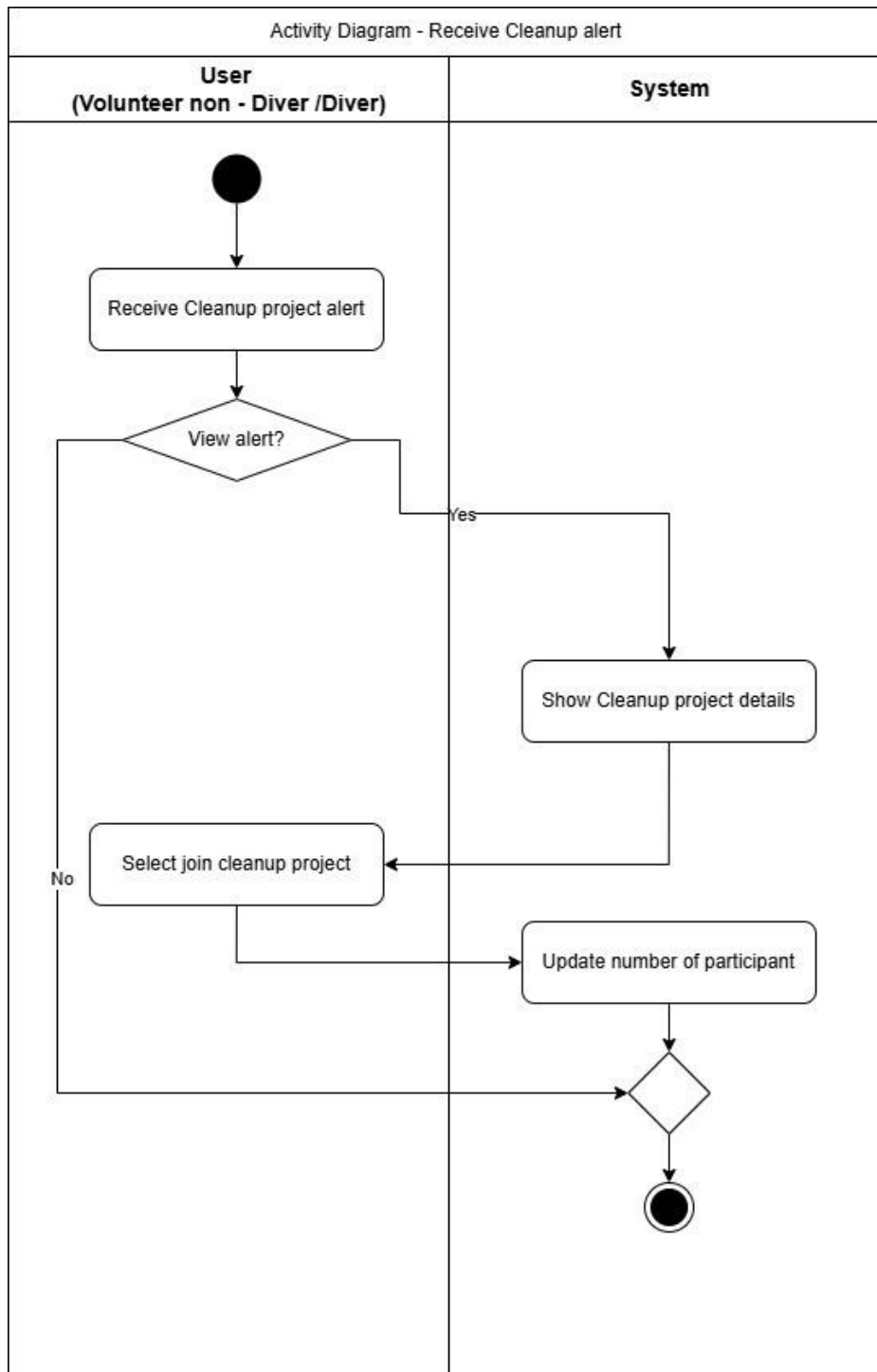


Figure 10: Activity Diagram - Alert and Notification Process

This activity diagram illustrates the process of responding to a cleanup project alert. The process begins when a volunteer non-diver or diver receives a cleanup project notification from the system. If the user chooses to view the alert, the system displays the project details, including information about the cleanup activity. The user can then decide to join the project, after which the system updates the number of registered participants. This process enables efficient volunteer participation and helps the system track involvement in cleanup activities.

Project Progress Monitoring

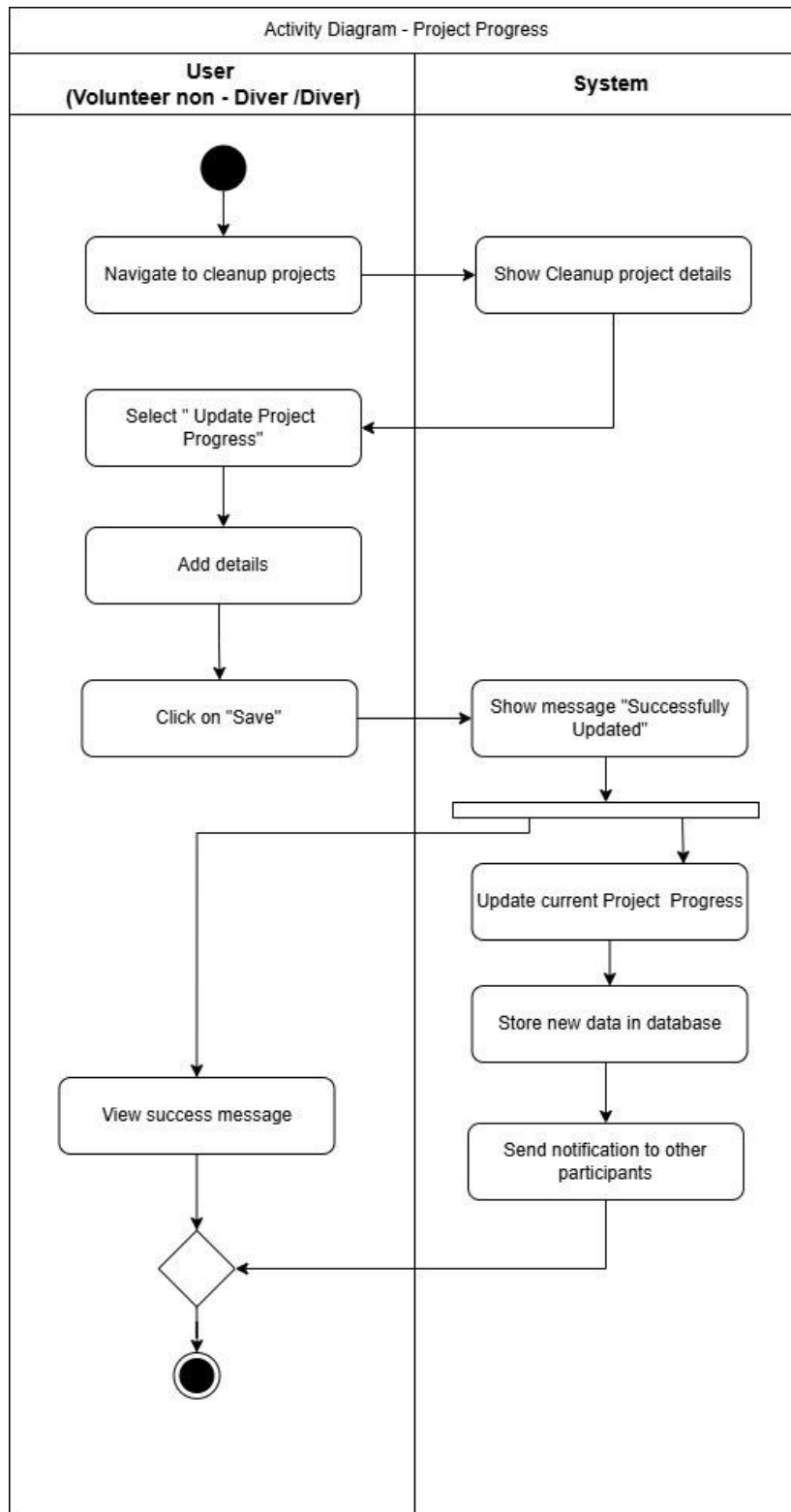


Figure 11: Activity Diagram - Project Progress Monitoring

This activity diagram illustrates the process of updating the progress of a cleanup project. The process begins when a user navigates to a cleanup project and selects the "Update Project Progress" option. The user then enters the required progress details and submits the update. The system displays a confirmation message, updates the current project progress, stores the new information in the database, and notifies other project participants about the update. This process ensures that project activities are accurately tracked and communicated to all relevant stakeholders.

6.2.3. State Chart Diagram Design

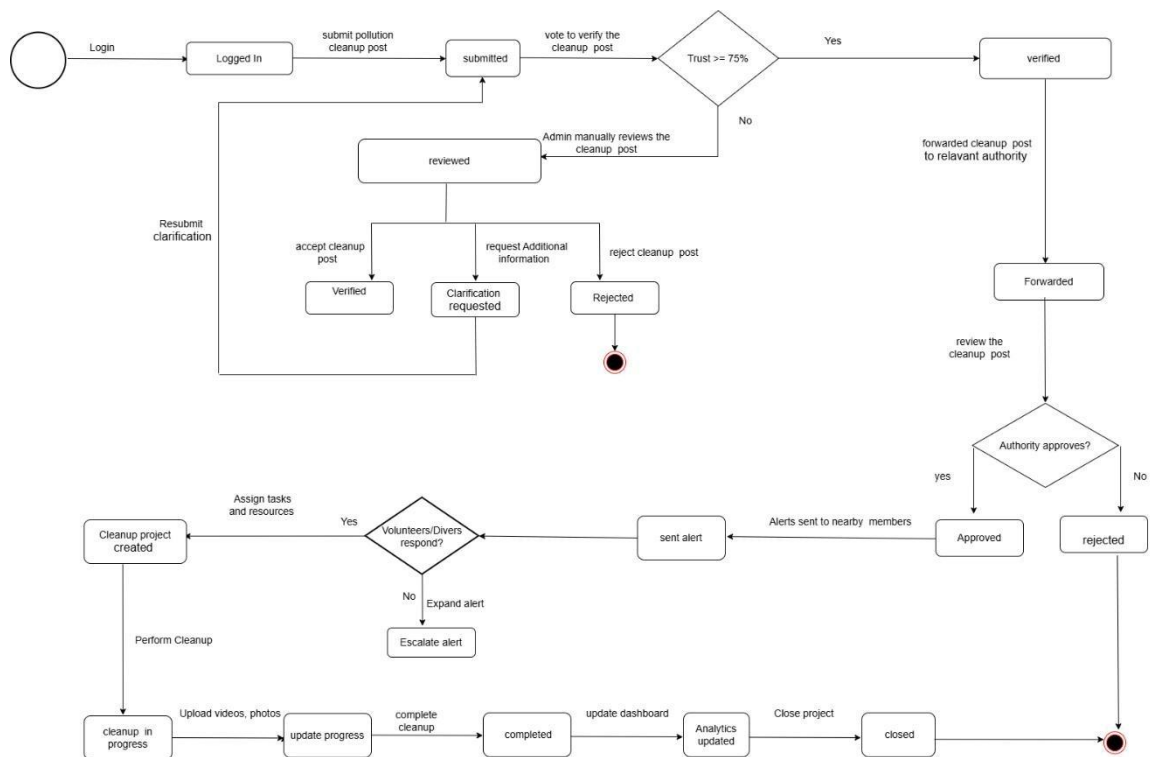


Figure 12: State Chart Diagram

This state chart diagram illustrates the lifecycle of a pollution cleanup post from submission to project completion. The process begins when a user logs into the system and submits a pollution

cleanup post. The post then enters the *Submitted* state and undergoes community verification, where a trust percentage is calculated based on user votes. If the required trust threshold is achieved, the post is marked as *Verified* and forwarded to the relevant government authority. If the trust threshold is not met, the post enters the *Reviewed* state, where an administrator may verify the report, request additional information, or reject it. Once approved by the relevant authority, alerts are sent to nearby volunteers and divers to encourage participation. After sufficient participant responses are received, a cleanup project is created and progresses through the states of *In Progress*, *Completed*, and *Closed*. Throughout the process, project updates, notifications, and analytics are generated to support monitoring and decision-making. This diagram demonstrates how the system manages the various states and transitions associated with pollution reporting, verification, approval, cleanup execution, and project closure.

6.2.4. Sequence Diagram Design

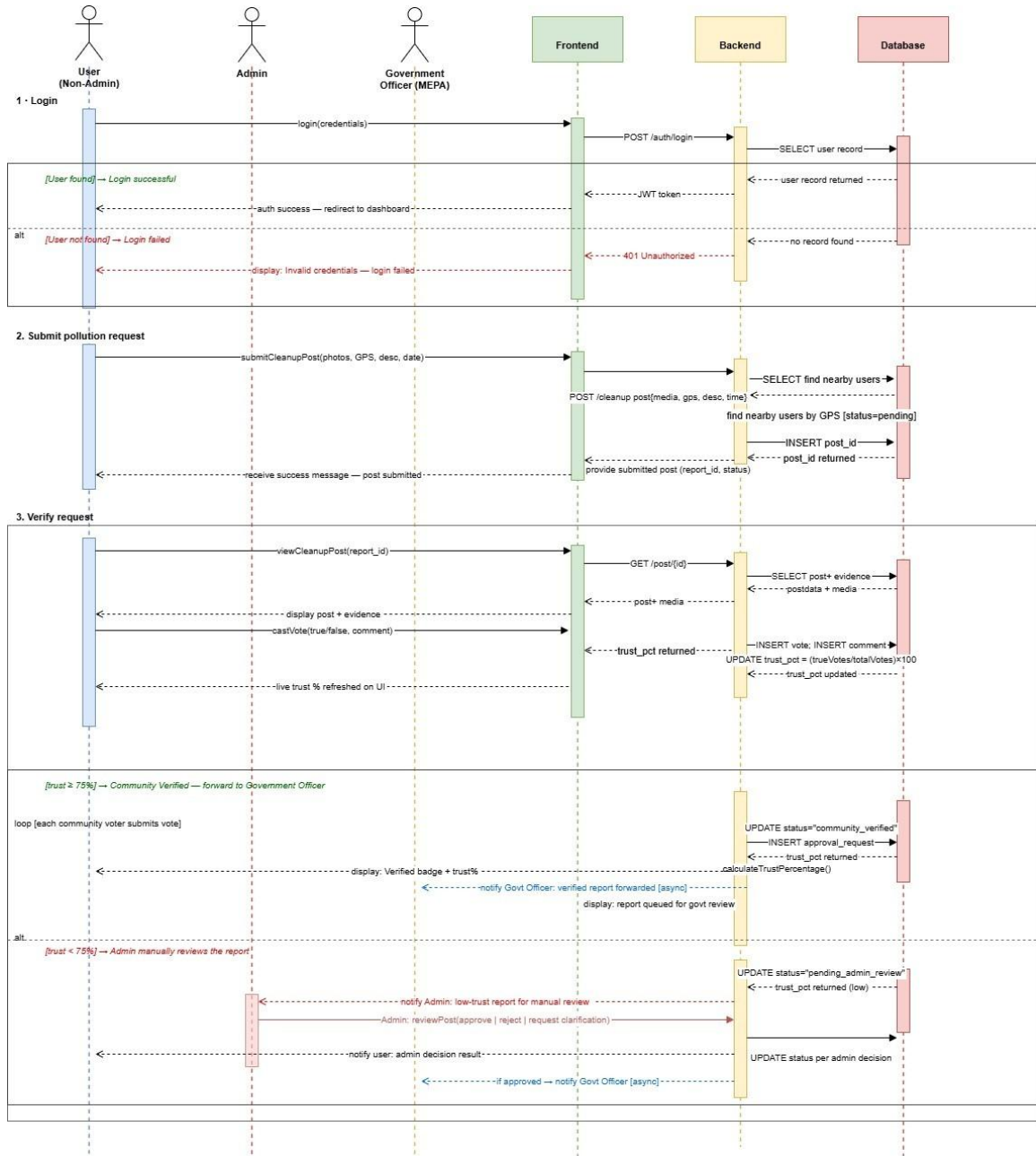


Figure 13: Sequence Diagram

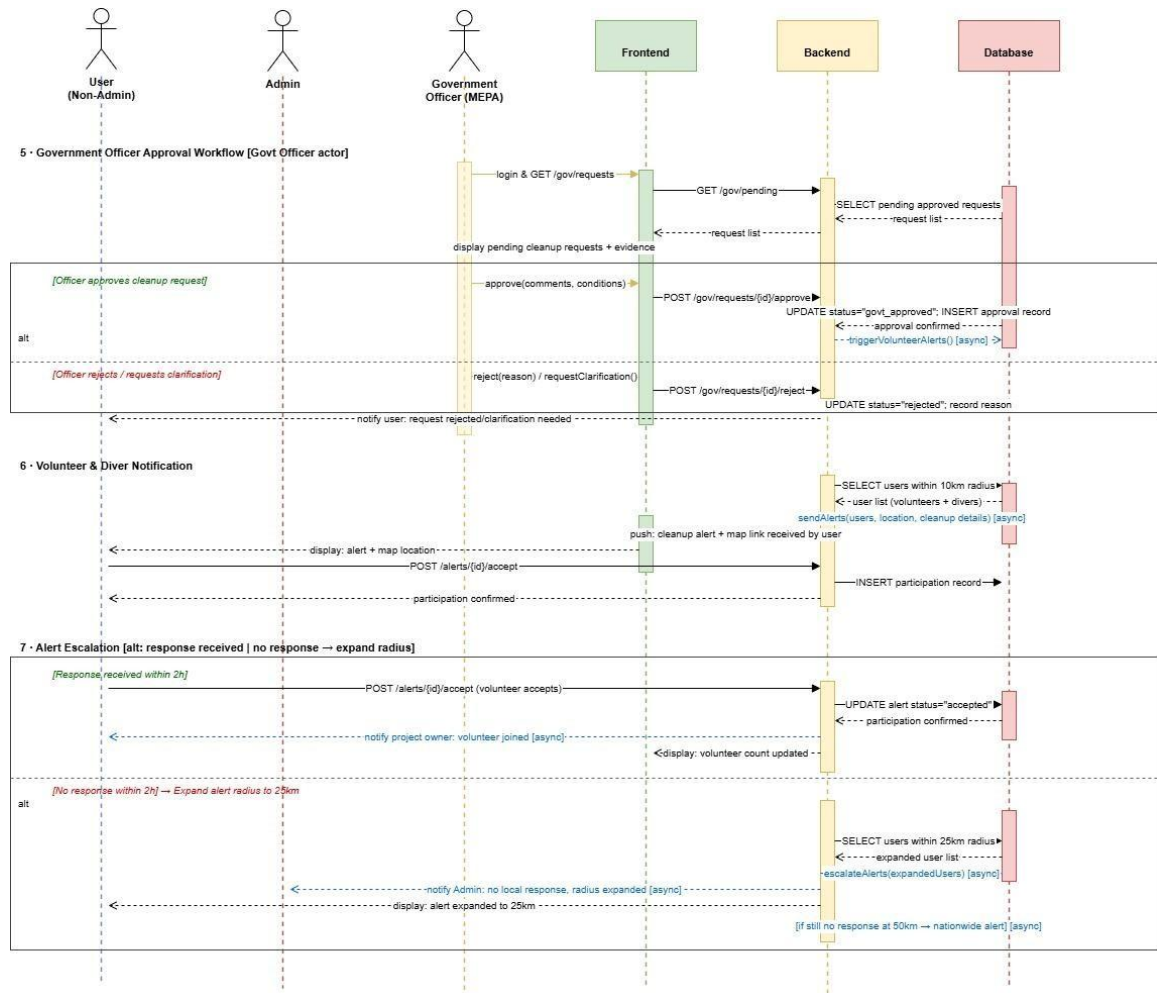


Figure 14: Sequence Diagram

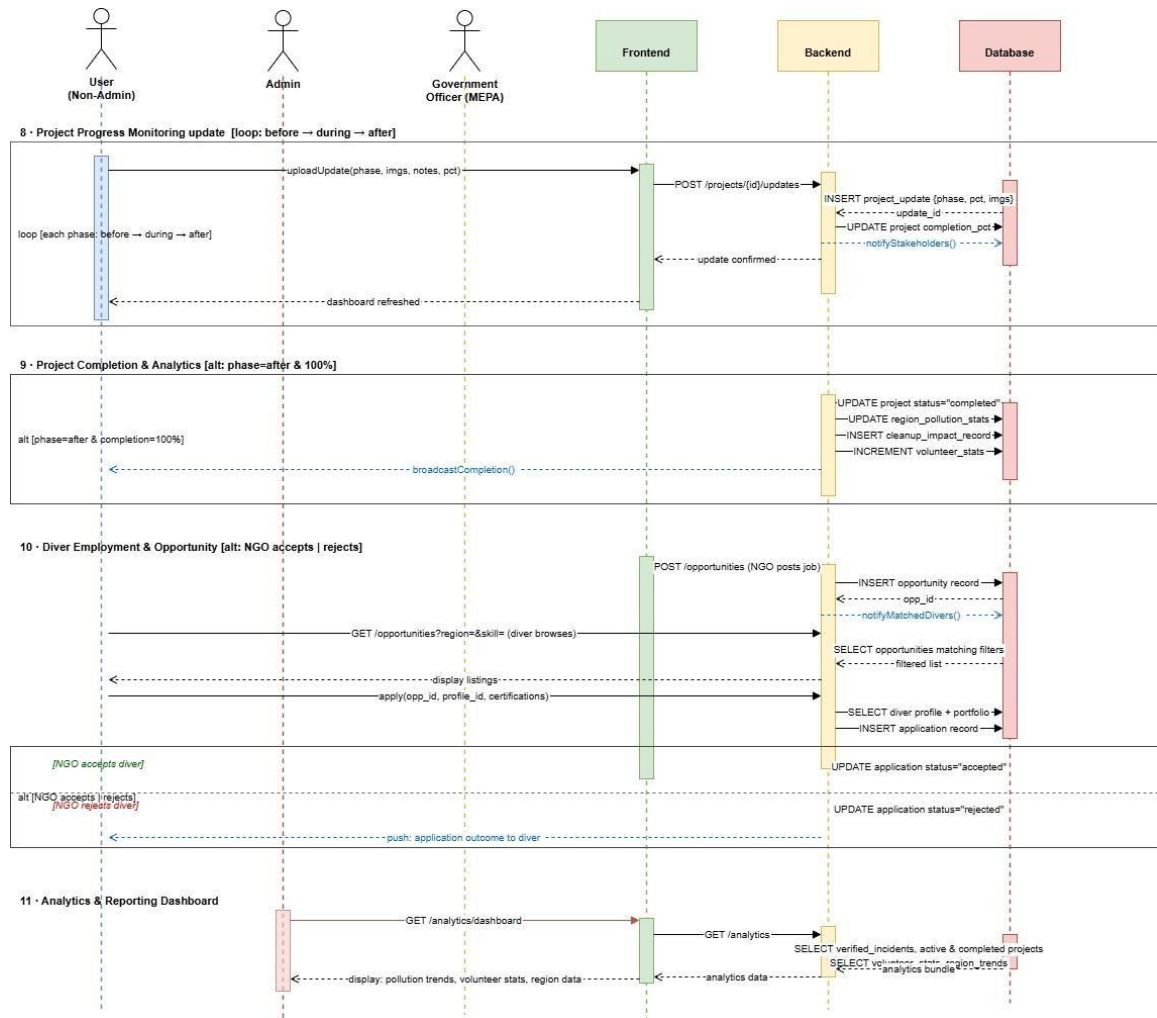


Figure 15: Sequence Diagram

This sequence diagram illustrates the interactions between users, administrators, government officers, and the system components during the execution of the major system workflows. The diagram shows how requests are exchanged between the frontend, backend, and database when users log in, submit pollution reports, participate in community verification, and receive notifications. It also demonstrates the administrative review process, government approval workflow, volunteer and diver alert management, project progress monitoring, diver opportunity management, and analytics generation. The sequence of messages highlights how data flows through the system and how different components collaborate to support pollution reporting, cleanup project coordination, approval processes, and environmental monitoring activities.

6.2.5. Class Diagram Design

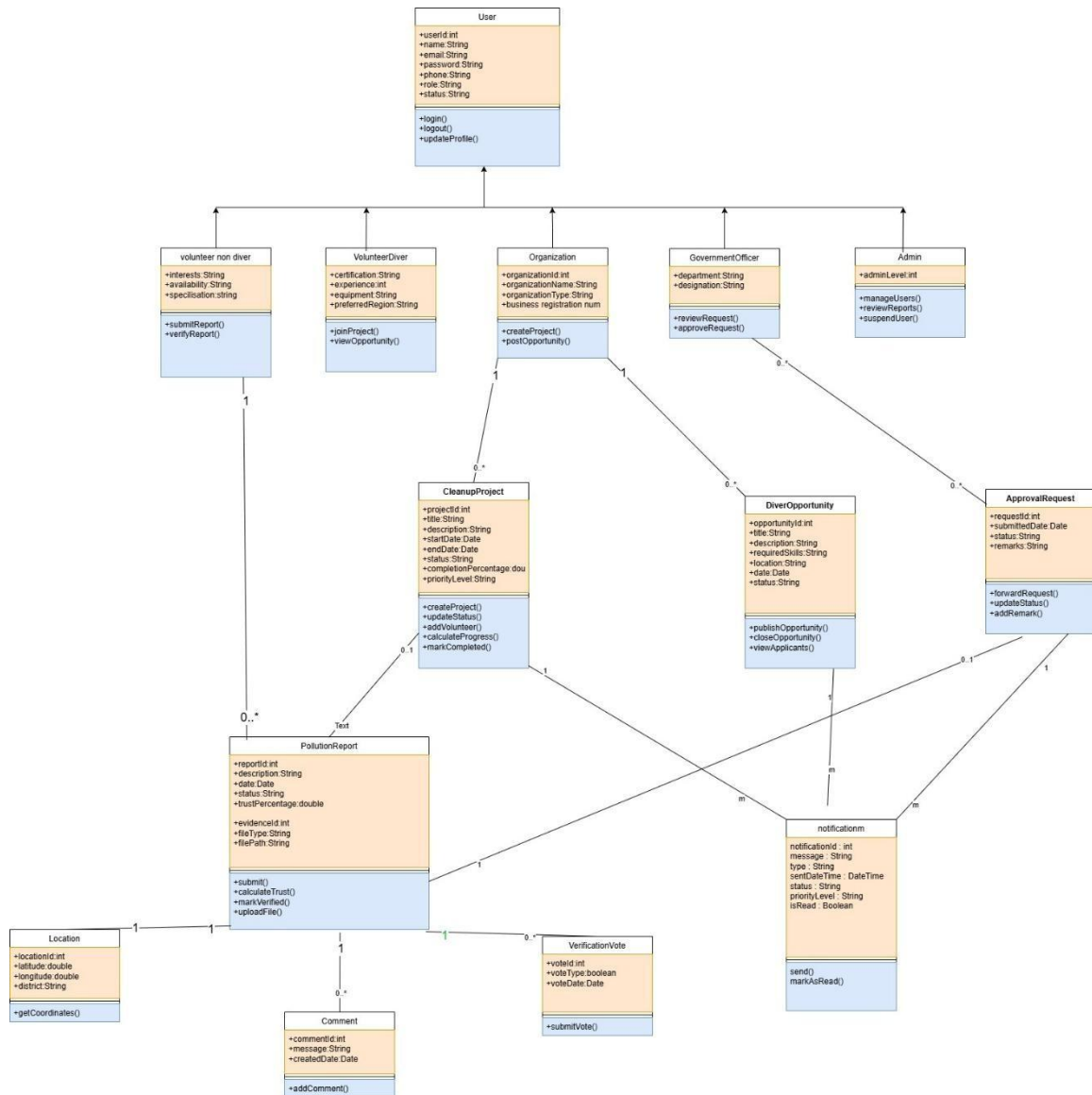


Figure 16: Class Diagram

This class diagram illustrates the static structure of the Community-Based Ocean & Coastal Cleanup Management System by showing the main classes, their attributes, methods, and relationships. The User class serves as the parent class, from which specialized user roles such as Volunteer Non-Diver, Volunteer Diver, Organization, Government Officer, and Admin inherit common properties and functionalities. The diagram also includes key system classes such as PollutionReport, Location, VerificationVote, Comment, CleanupProject, DiverOpportunity,

ApprovalRequest, and Notification, which support the core operations of the system. The relationships among these classes demonstrate how users submit and verify pollution reports, participate in cleanup projects, manage diver opportunities, process approval requests, and receive notifications. Overall, the class diagram provides a detailed representation of the system's object-oriented design and supports the implementation of the functional requirements identified for the proposed solution.

6.2.6. ER Diagram Design

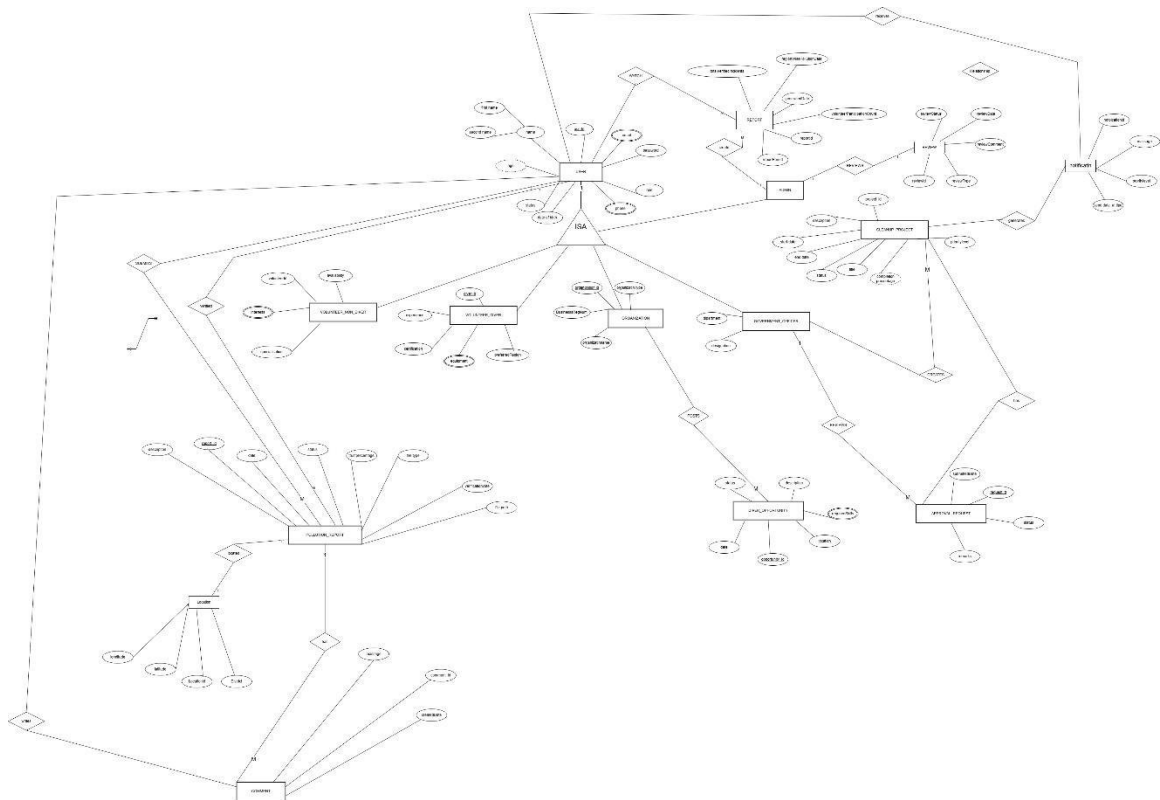


Figure 17: ER Diagram

This Entity Relationship (ER) Diagram illustrates the database structure of the Community Based Ocean & Coastal Cleanup Management System and the relationships among its major entities. The diagram includes core entities such as User, Volunteer Non-Diver, Volunteer Diver, Organization, Government Officer, Pollution Report, Comment, Report, Review, Approval Request, Cleanup Project, Diver Opportunity, Notification, and Location. It demonstrates how users submit pollution

reports, participate in verification activities, create and manage cleanup projects, request approvals, post diver opportunities, and receive notifications. The relationships between these entities ensure efficient data management and support the system's key functionalities, including pollution reporting, community verification, government approval workflows, volunteer coordination, project monitoring, and environmental analytics.

6.3. Summary

This chapter presents the design of the Community-Based Ocean & Coastal Cleanup Management System through a series of design diagrams and models. The use case, activity, sequence, class, state chart and ER diagrams were used to illustrate the structure, behaviour, and interactions of the system. These design artifacts provide a clear representation of the system architecture, workflows, database structure, and major components, ensuring that the identified requirements are effectively translated into a practical and implementable solution. Together, they serve as the foundation for the development and operation of the proposed system.

7. Conclusion

The Community-Based Ocean & Coastal Cleanup Management System was proposed to address challenges associated with marine pollution reporting, volunteer coordination, and cleanup project management. Traditional reporting methods and social media-based communication platforms often lack structured workflows, verification mechanisms, and effective coordination capabilities, making large-scale environmental cleanup activities difficult to manage.

To overcome these limitations, the proposed system provides a centralized Progressive Web Application (PWA) that supports pollution reporting, community verification, government approval workflows, volunteer and diver coordination, project monitoring, and analytics generation. By integrating these functionalities into a single platform, the system improves collaboration among community members, environmental organizations, and government authorities.

The use of modern technologies such as React.js, Spring Boot, and MySQL enables the development of a secure, scalable, and user-friendly solution. Furthermore, features such as location-based notifications, project progress tracking, and analytical reporting contribute to more effective environmental monitoring and decision-making. Overall, the proposed system offers an integrated approach to supporting ocean and coastal cleanup initiatives while encouraging greater community participation in environmental conservation activities.

7.1. Future Improvements

Although the proposed system provides a comprehensive platform for pollution reporting, volunteer coordination, and cleanup project management, several enhancements can be incorporated in future versions to further improve its functionality and effectiveness.

Potential improvements include:

- Implementation of AI-based pollution detection and classification using image processing and machine learning techniques.

- Integration of real-time GPS tracking to monitor volunteer participation and project activities.
- Integration with government environmental databases and information systems for improved data sharing and collaboration.
- Support for multiple languages to accommodate a wider range of users and communities.
- Addition of real-time communication and messaging features to improve coordination among volunteers, organizations, and authorities.
- Development of advanced analytics, predictive models, and environmental trend forecasting capabilities to support decision-making and resource planning.
- Enhancement of the PWA with advanced push notifications and background synchronization to provide timely project updates, alerts, and automatic synchronization of offline data when an internet connection becomes available.

These enhancements would further strengthen the system's ability to support environmental conservation initiatives and promote greater community participation in ocean and coastal cleanup management activities.

REFERENCES

- [1] United Nations Environment Programme (UNEP), “Marine Plastic Pollution,” UNEP, 2024.
- [2] Marine Environment Protection Authority (MEPA), Sri Lanka, “Beach Caretaker Program,” 2025.
- [3] Marine Debris Tracker, “Marine Debris Tracker,” Available: <https://marinedebris.engr.uga.edu>
- [4] Ocean Conservancy, “Clean Swell,” Available: <https://oceanconservancy.org>
- [5] Parley Sri Lanka, “Parley Sri Lanka,” Available: <https://www.parley.tv>
- [6] React Documentation, Available: <https://react.dev>
- [7] Spring Boot Documentation, Available: <https://spring.io/projects/spring-boot>
- [8] Oracle, “Java Documentation,” Available: <https://docs.oracle.com>
- [9] MySQL Documentation, Available: <https://dev.mysql.com/doc>
- [10] Google Maps Platform Documentation, Available: <https://developers.google.com/maps>
- [11] Docker Documentation, Available: <https://docs.docker.com>
- [12] Apache Maven Documentation, Available: <https://maven.apache.org>
- [13] Apache Tomcat Documentation, Available: <https://tomcat.apache.org>